Scalable Analysis of Vegetation Anomalies Preceding Plant Disease Outbreaks

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1 Motivation

The motivation for this project is mainly practical. Plant diseases can cause long-term ecological damage and economic loss if they are not identified in a timely manner. In agricultural settings, farmers and local communities can often observe visible signs of disease and take corrective action. However, in forested and remote regions, such direct monitoring is limited or nonexistent, allowing diseases to spread undetected for extended periods. Analyzing vegetation behavior over time using data-driven methods can provide valuable insight into early stress patterns that may otherwise go unnoticed. In addition, this project offers a secondary technical motivation by enabling the application of big-data concepts to a real-world environmental problem. However, the emphasis remains on understanding and interpreting vegetation changes rather than guaranteeing predictive accuracy.

2 Related Work

Researchers have conducted extensive studies about forest disturbance detection through satellite imaging which requires three specific methodologies: time series modeling and spectral analysis and unsupervised learning methods. The Sentinel-2 and Landsat sensors deliver continuous multi-spectral time series data which enables them to detect minor vegetation stress before it becomes visible to the human eye.

Recent systems have implemented cloud-native formats through their adoption of Cloud Optimized GeoTIFFs (COGs) and PostgreSQL spatial databases combined with PostGIS. The technologies provide users with two key functions which allow them to filter spatial data across different time periods while accessing extensive geospatial information.

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Microsoft's Planetary Computer and Radiant Earth's STAC catalog demonstrate this systematized framework through their respective projects.

The project develops a spatiotemporal data cube through COG-based raster storage and PostGIS metadata indexing and the PySpark ETL pipeline. The system enables organizations to detect anomalies at scale through its combination of traditional and contemporary detection methods.

3 Data Sources and Study Context

3.1 Study Region and Timeline

This study focuses on Boulder County, Colorado, a region significantly impacted by the Mountain Pine Beetle (MPB) outbreak. The MPB infestation has caused widespread forest mortality across Colorado, making it an appropriate case study for analyzing vegetation stress preceding documented disease events.

- **Region:** Boulder County, Colorado
- **Disease Focus:** Mountain Pine Beetle (MPB)
- **Temporal Scope:** 2015–2025 (10-year archive)

The selected timeline enables construction of seasonal vegetation baselines and retrospective analysis of anomalous behavior prior to documented outbreaks.

3.2 Satellite Imagery: Harmonized Landsat–Sentinel-2 (HLS)

The primary satellite dataset used is the Harmonized Landsat–Sentinel-2 (HLS) surface reflectance product, which integrates Landsat and Sentinel-2 imagery into a consistent multispectral dataset.

- Provides standardized surface reflectance data
- Includes visible, near-infrared, and shortwave infrared bands
- Supports computation of NDVI and NDMI
- High temporal frequency suitable for time-series analysis

These properties make HLS suitable for detecting subtle vegetation stress patterns over time.
For the Boulder County study area (approximately 4,900 km^2), HLS imagery at 30 m resolution results in roughly 5–6 million pixels per acquisition. After retaining 6–8 spectral bands and accounting for quality layers and file overhead, each AOI-clipped scene is estimated to be approximately 80–110 MB. With roughly 800–1100 usable scenes over the 2015–2025 period, the projected total dataset size is approximately 65–120 GB, depending on band selection and quality filtering.

3.3 Land Cover Data: ESA WorldCover 10 m

The ESA WorldCover 10 m global land cover dataset is used to isolate forested areas within the study region.

- 10 m spatial resolution land cover classification
- Used to apply a forest mask
- Reduces noise from cropland, grassland, and urban surfaces

Restricting analysis to forest pixels improves ecological relevance and anomaly interpretation.

3.4 Data Access via Google Earth Engine

Satellite data retrieval and preliminary filtering are conducted using Google Earth Engine (GEE).

- Cloud-based access to large satellite archives
- Filtering by region (Boulder County)
- Filtering by time range (2015–2025)
- Quality filtering (e.g., cloud coverage)

4 Data Management and Indexing

4.1 Raw Data Storage Layer

- **Raster Data:** All satellite imagery (HLS scenes) is stored as Cloud Optimized GeoTIFFs (COGs).
  - Located in cloud object storage (AWS S3).
  - COGs are optimized for fast, partial reads and cloud-native processing.
- **Metadata:** Stored separately in a PostgreSQL database with PostGIS for efficient querying.
  - Includes spatial (bounding box), temporal (acquisition date), and descriptive (band, cloud cover) attributes.

4.2 Batch Ingestion (ETL Pipeline)

A PySpark + Rasterio pipeline processes raw HLS files. For each image:

(1) Traverse directories or cloud buckets.
(2) Extract metadata (date, tile ID, band, cloud cover, bounding box).
(3) Convert to COG format if needed.
(4) Store metadata in PostgreSQL.

4.3 Spatial Indexing (PostGIS)

Bounding box is read from each COG using Rasterio.

- A geometry is created using `ST_MakeEnvelope(...)` in PostGIS.
- A spatial index is built using `CREATE INDEX ... USING GIST`.
- Enables fast spatial queries like `ST_Intersects` and `ST_Contains`.

4.4 Spatiotemporal Data Cube Index

Final output: a centralized metadata table that links:

- **Space** → Bounding box geometry.
- **Time** → Acquisition date.
- **Link** → File path or S3 URL to the COG.

Stored in PostgreSQL with PostGIS indexing. Queried using SQL for spatiotemporal filtering. This centralized index serves as an entry point for the downstream data transformation pipeline, enabling the efficient retrieval of filtered pixel-level reflectance data for feature computation as described in Section 5.

5 Data Transformation

When a user submits a query for a specific time range, the system dynamically retrieves the relevant data by first identifying candidate scenes using the spatial indices. It then uses the metadata in PostgreSQL/PostGIS to perform partial reads of the Cloud Optimized GeoTIFFs (COGs) stored in S3, fetching only the required spectral bands (Red, NIR, and SWIR).

5.1 Forest Mask Application and Quality Filtering

- **Data Source:** Uses ESA WorldCover 10m land cover map.
- **Method:** Apache Sedona spatial join between HLS pixel centroids and WorldCover polygons.
- **Filtering:** Removes non-forest pixels (cropland, grassland, water, built-up surfaces) to reduce computational load.

5.2 Vegetation Index Derivation

Computes spectral indices from Harmonized Landsat–Sentinel-2 (HLS) data using PySpark.

- **Indices:** NDVI (Normalized Difference Vegetation Index) and NDMI (Normalized Difference Moisture Index).

$$NDVI = \frac{NIR - Red}{NIR + Red}, \quad NDMI = \frac{NIR - SWIR1}{NIR + SWIR1} \tag{1}$$

  (Derived from Red, Near-Infrared (NIR), and Shortwave Infrared (SWIR) bands).

5.3 Temporal Harmonization and Gap Handling

Standardizes irregular satellite observations into a consistent, regular temporal sequence.

- **Aggregation:** Monthly composites to smooth noise and ensure comparability.
- **Gap Filling:** Short-gap interpolation for missing months.

6 Query-Triggered Anomaly detection

To ensure the system remains responsive and memory efficient, the transformation and detection logic are not pre calculated and are instead executed on the fly for the specific spatiotemporal window requested by the user.

6.1 Standardized Z-Score Anomaly Logic

Immediately following index derivation, the anomaly detection logic is applied to the filtered dataset to identify significant vegetation stress.

- **Statistical Normalization:** For every pixel in the user's query timeframe, the observed index value is compared against a pre-computed historical baseline (2015–2025). The Z-score is calculated as:

$$Z = \frac{\text{Observed Value} - \mu_{monthly}}{\sigma_{monthly}} \tag{2}$$

  where $\mu_{monthly}$ and $\sigma_{monthly}$ are the historical mean and standard deviation of the selected vegetation index value (such as NDVI or NDMI) for that specific pixel and month.
- **Anomaly Thresholding:** A pixel is flagged as an anomaly if $Z < -2.0$, representing observations in the lower tail of the distribution more than two standard deviations below the historical mean, and thus indicating unusually low vegetation index values for that season.

- **Query Output:** The system returns the results as either a *Raster Anomaly Map* for spatial visualization or a *Tabular Event Dataset* in Parquet format, providing immediate feedback for the user's specific query.

7 Evaluation Plan

The project will be evaluated along three primary dimensions to assess both computational performance in a Big Data environment and the analytical rigor of early vegetation anomaly detection.

7.1 Scalability Evaluation (Big Data Performance)

To evaluate scalability, the full pipeline will be executed on progressively increasing data volumes (e.g., 3 years, 5 years, and 10 years of satellite imagery). Apache Spark metrics will be used to measure system behavior under increasing workload.

- Processing throughput (e.g., spatial tiles or GB processed per minute)
- Total runtime as a function of data volume
- Peak executor memory usage

Scalability is demonstrated if throughput remains stable and total runtime grows approximately linearly with increasing data size, without excessive memory pressure or disk spill.

7.2 Concurrency Evaluation (Parallel Processing Efficiency)

Concurrency is evaluated by examining stages with naturally parallel workloads, such as ingestion, preprocessing, and NDVI/NDMI feature computation across spatial tiles.

Apache Spark metrics will be used to measure:

- Number of active tasks per stage (parallel tile and time-series computations)
- Executor CPU utilization during parallel stages

Efficient concurrency is demonstrated if multiple spatial tiles are processed simultaneously across executors and CPU utilization remains consistently high, indicating effective use of distributed computing resources.

7.3 Query Efficiency Evaluation

To assess data management efficiency, representative spatiotemporal queries (e.g., region-based queries over varying temporal ranges of satellite imagery) will be executed.

Each query will be run both with spatial indexing enabled and without spatial indexing to compare performance.

Apache Spark and Apache Sedona metrics will be used to measure:

- Query execution time (latency)
- Number of spatial tiles or records scanned per query
- I/O cost during query execution

Query efficiency is demonstrated if indexed queries exhibit significantly lower latency and reduced data scans compared to non-indexed queries, and if query time increases gradually rather than sharply as the queried temporal range grows.

7.4 Statistical Model Evaluation (Anomaly Classification Performance)

To evaluate the effectiveness of the anomaly detection component, the problem is formulated as a binary classification task in which each pixel is labeled as either anomalous or normal based on the Z-score thresholding criterion.

The predicted class labels are derived directly from the statistical test, where a pixel is classified as an anomaly if its computed Z-score falls below the predefined threshold. Model performance is assessed by comparing these predicted labels against reference anomaly annotations when available.

The following standard classification metrics will be used:

- Accuracy: Overall proportion of correctly classified pixels.
- Precision: Proportion of predicted anomalies that are true anomalies.
- Recall: Proportion of true anomalies that are successfully detected.

The statistical model is considered effective if it achieves high recall while maintaining reasonable precision, ensuring that vegetation stress events are detected reliably without excessive false alarms.

8 Output Products and Visualization

8.1 Forest Health Visualization

We will develop a simple web tool which allows users to select timeframes for the given region to run the anomaly detection pipeline on. The dashboard provides two primary visual outputs:

(1) **Anomaly Information**: Displays real-time Z-scores and a Raster Anomaly Map for spatial visualization or a Tabular Event Dataset in Parquet format.
(2) **Spectral Timelapse**: Renders an animated sequence of satellite imagery (RGB) for the selected timeframe, enabling possible visual confirmation of canopy stress or disturbance events.

9 Conclusion

This project proposes a scalable framework for detecting vegetation anomalies preceding plant disease outbreaks using large-scale satellite time-series data. Focusing on Boulder County, Colorado and the Mountain Pine Beetle (MPB) case study, the system integrates Harmonized Landsat–Sentinel-2 imagery, ESA WorldCover forest masking, cloud-native raster storage, and distributed PySpark processing to construct a spatiotemporal analysis pipeline. By combining efficient indexing, on-demand query execution, and Z-score–based anomaly detection, the architecture demonstrates how interpretable statistical methods can be deployed effectively within a Big Data environment.

Through structured evaluation of scalability, concurrency, query efficiency, and anomaly classification performance, the project validates both computational robustness and analytical rigor. Overall, this work demonstrates how scalable data engineering and ecological analysis can be combined to support large-scale forest health monitoring and future system expansion.

10 Author Contributions

This section specifies the responsibilities and work performed by each team member for this literature review deliverable.

- **Thrisha Ambareesharaje Urs Urs**: Wrote the Data Sources and Study Context section, detailing the study region, timeline, and dataset characteristics. Authored the Conclusion section summarizing the overall system design and evaluation strategy. Performed final document review to ensure consistency and alignment across all sections.
- **Sohum Damani**: Looked into data storage and indexing part. I update the related work to match our tech stack and find relevant indexing practices that can be used in our project. Discussed with the team about evaluation plan and how we can display our results.
- **Yashaswini Diggavi**: I looked into the evaluation plan and discussed with the team how the system would be evaluated. I then updated the report by adding evaluations for scalability, concurrency and query efficiency.
- **Vedant Borkute**: Worked on data preprocessing and transformations (Section 5), as well as output products (Section 8). Discussions on the evaluation plan & the project pipeline.
- **Shreyangshu Bera**: Wrote the anomaly detection section and contributed the corresponding evaluation methodology based on classification metrics derived from the Z-score analysis.

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